From Cancellation Risk to Revenue Decisions: A Hybrid, Probability-Aware Hotel Overbooking System

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***Abstract*—Hotel cancellations introduce uncertainty in room demand, resulting in unused inventory and revenue loss. This research develops a cancellation-aware hotel overbooking framework that predicts booking cancellations, calibrates probabilities, and translates them into show probabilities for overbooking decisions. The proposed methodology evaluates machine learning models, probability calibration, and risk-adaptive policies by assessing profit, walked guests, and unused rooms across varying capacity and cost scenarios.**

***Index Terms*—Machine Learning, Hotel Overbooking, Cancellation Prediction, Probability Calibration, Revenue Management.**

I. Introduction

The primary issue examined in this paper is hotel overbooking in the presence of cancellation uncertainty. The following section outlines the problem's context, its relevance to hotel revenue management, and the rationale for employing machine learning techniques. Addressing this challenge is significant due to its potential effects on occupancy, profitability, service quality, and operational decision-making.

A. Scope and Objectives

This project seeks to develop a cancellation-aware hotel overbooking decision system. The objectives are as follows:

- Establish the relevance of hotel overbooking to revenue management.
- Demonstrate originality through calibrated, risk-aware decision-making.
- Achieve robust results through machine learning and optimization techniques.
- Ensure reproducibility and transparency in data, methodology, and evaluation.

II. Related Work

This section examines existing literature and prior research concerning hotel cancellation prediction and overbooking. Previous studies have explored cancellation prediction, probability calibration, and revenue management, yet they vary in the extent to which these methods are integrated. The present approach extends prior work by incorporating calibrated cancellation probabilities into risk-aware overbooking strategies, thereby enhancing originality and relevance within the hotel business context.

III. Methodology

This section details the machine learning methodology, including data sources, preprocessing steps, model selection, and evaluation metrics, designed to ensure reproducibility and technical rigor.

A. Data

The analysis utilizes a hotel reservations dataset obtained from Kaggle. Preprocessing procedures involve date validation and feature engineering. This approach enhances transparency and reproducibility.

B. Models and Techniques

The machine learning models applied are Logistic Regression, Random Forest, and Gradient Boosting. Model selection is based on comparative performance. Calibration and optimization procedures further support reproducibility.

C. Evaluation Metrics

The evaluation metrics include Accuracy, F1-Score, ROC-AUC, Brier Score, and Log Loss. These metrics are selected to align with business objectives and to provide a comprehensive assessment of model performance.

IV. Results

ML improves decisions under favorable risk conditions, but the historical policy remains competitive when guest-walk costs are high, supporting a risk-adaptive overbooking strategy. Results are presented using tables and visualizations, as shown in Table I and Figure IV.

V. Discussion

The findings indicate that the hybrid forecasting method improves probability accuracy and enables better economic decisions when compared with a policy of no overbooking. It is shown that Gradient Boosting provides good predictive performance and that combining calibrated probabilities with

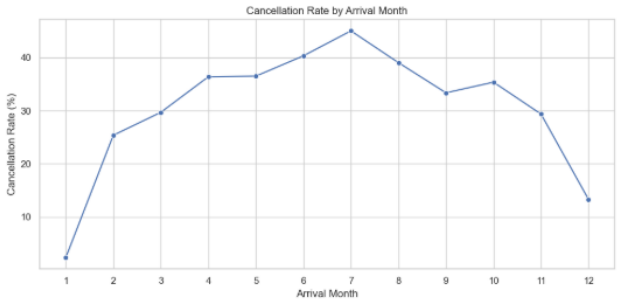


Fig. 1. Visualization of Model Performance

historical cancellation data increases the robustness of the approach. The business relevance is illustrated by an examination of the trade-offs between profit and service risk in various scenarios. The originality of the study is due to the application of hybrid forecasting and Poisson-binomial optimization, and the methodological soundness is supported by a chronological assessment. The paper clearly recognizes its limitations in order to assist in the interpretation of the results

VI. Prompts to Enhance AMLBT Project or Paper

The following prompts are being provided in order to enhance the quality and alignment of both the Advanced Machine Learning for Business Transformation (AMLBT) project and the associated paper with the evaluation criteria. Grouped according to the criteria of the rubric, these prompts are meant to help the team meet each requirement effectively and to ensure a solid, high-quality submission.

- **Relevance**
  - What is the best way of clearly setting out the hotel overbooking problem in order to show that it agrees with the aims of hotel revenue management?
  - Which business metrics, for example profit, occupancy, the number of empty rooms, and the number of guests who have been walked, will be improved by the proposed system?
  - What examples from the industry can we give to show that cancellation-aware overbooking is relevant to hotel managers?
- **Significance**
  - What could be the effect on the financial position and on the operations of hotels of improving their overbooking decisions?
  - What is the way of measuring the business value of calibrated cancellation probabilities and risk-adaptive booking recommendations?
- **Originality**
  - What is the way in which we can show that this project goes beyond ordinary hotel cancellation prediction by linking the predictions directly to overbooking decisions?
  - What is the way in which we can show that the risk-adaptive policy offers new managerial insights under various capacity and walking-cost scenarios?
- **Achievement**
  - What is the best way of clearly showing the results of the model and the business simulation to prove that successful execution has taken place?
  - Which tables and visualizations most effectively communicate calibration quality, simulated profit, the number of guests who have been shown around, and the number of rooms that have not been used?
  - What is the way in which we can examine the nine scenarios in order to understand why the hybrid policy wins in some of them while the historical policy is better in the others?
- **Writing**
  - What should we do to organize the paper so that it is clear, concise, and conforms to the IEEE conference formatting guidelines?
  - What strategies are there to make the discussion of prediction, calibration, optimization, and business impact more professional and concise?
  - What is the best way to effectively incorporate tables, figures, and equations without disrupting the flow of the paper?
- **Reproducibility**
  - What steps ought we to record in order that another researcher be able to reproduce the entire cancellation-prediction, calibration, and overbooking pipeline?
  - In order to make the code, the cleaned data, the assumptions, and the methodology transparent and accessible, how should we organize the GitHub repository?
  - What methods can we use to deal with limitations such as employing cancellation status as a proxy indicator, relying on scenario-based capacity, and using simulated walking costs?
  - Include the GitHub repository link in the paper.
- **Technical Quality**
  - What grounds there is for claiming that Logistic Regression, Random Forest, and Gradient Boosting are suitable for use in predicting hotel cancellations?
  - What makes the Brier Score, Log Loss, ECE, ROC-AUC, and PR-AUC suitable for evaluating cancellation probabilities and calibration?
  - What grounds there is for using calibrated show probabilities and the Poisson-binomial distribution in order to optimise overbooking?
- **Responsible Use of Prompts**
  - They should be used to refine the ideas and to improve alignment with the AMLBT rubric, not for generating the whole paper or code.

- AI-generated content should be reviewed and edited to make sure that it is accurate, relevant, and in line with the real project results.
- A note regarding any AI assistance used should be included in the AI Use Declaration section.

VII. Literature Review and Bibliography Guidelines

This section outlines the steps for conducting a literature review and creating a bibliography in IEEE format to ensure a robust Related Work section (Section II) that supports the project's relevance, significance, and originality.

A. Conducting a Literature Review

To perform an effective literature review for Your Business Problem:

- **Identify Relevant Sources**: Search academic databases such as IEEE Xplore, Google Scholar, and Scopus using keywords related to hotel booking cancellation, overbooking, cancellation prediction, probability calibration, revenue management, and machine learning. Focus on peer-reviewed research and recent studies in the hotel revenue-management domain.
- **Evaluate Source Quality**: Prioritize peer-reviewed journals and reputable conference proceedings that address hotel cancellation prediction, no-show forecasting, probability calibration, and overbooking optimization. Assess the relevance, credibility, methodology, and applicability of each source to the proposed hotel overbooking problem.
- **Synthesize Findings**: Summarize Key findings, methodologies, and limitations of prior studies were summarized. The literature was grouped into themes, including hotel cancellation prediction, machine learning models, probability calibration, no-show probability forecasting, and overbooking optimization, to provide a structured understanding of existing research.
- **Position Your Work**: The proposed approach addresses gaps in existing research by connecting calibrated cancellation probabilities directly to overbooking decisions. The methodology incorporates a 55% calibrated machine-learning and 45% historical hybrid, Poisson-binomial optimization, and risk-adaptive evaluation across various capacity and walking-cost scenarios to clarify the project's contribution.
- **Cite Appropriately**: All in-text citations were formatted using IEEE citation style, and every referenced academic work was included in the bibliography. Key references include research on hotel overbooking based on no-show probability forecasts and machine-learning-based hotel cancellation prediction.

B. Creating an IEEE-Format Bibliography

The bibliography must follow IEEE format, as implemented in this template using the `thebibliography` environment and `bibliographystyle{IEEEtran}`. Guidelines include:

- **Format Entries Correctly**: Each entry should include author(s), title, journal/conference name, volume, issue, pages, and year, formatted as shown in the References.
- **Order Citations Numerically**: Citations are numbered in the order they appear in the text (e.g., [1], [2]). Use `\cite{label}` for in-text citations.
- **Use Reliable Sources**: Include peer-reviewed or reputable sources relevant to hotel cancellation prediction, probability calibration, hotel overbooking, and revenue management. The Kaggle dataset may be included as a data source, but it should not replace scholarly references.
- **Manage References Efficiently**: Use a reference management tool (e.g., Zotero, Mendeley, Google Scholar) to organize sources and generate IEEE-compliant BibTeX entries, which can be adapted for the `thebibliography` environment.
- **Keep Within Page Limit**: Ensure the bibliography fits within the 4-page limit, prioritizing the most relevant references (typically 5–15 entries for a conference paper).

C. Table Example

A simple table can be used to present data or results clearly. Below is an example of a table comparing model performance metrics.

TABLE I
Example Model Performance Comparison

Model	ROC-AUC	PR-AUC	Brier	Log Loss
Logistic Regression	0.718	0.628	0.218	0.625
Random Forest	0.726	0.625	0.208	0.597
Gradient Boosting	0.765	0.727	0.199	0.575

D. List Example

Lists are useful for summarizing steps, objectives, or findings. Below is an example of a bulleted list in IEEE format.

- Validate and preprocess the hotel reservation data.
- Train Logistic Regression, Random Forest, and Gradient Boosting models.
- Calibrate cancellation probabilities using isotonic regression.
- Convert cancellation probabilities into show probabilities.
- Aggregate show probabilities using the Poisson-binomial distribution.
- overbooking decisions under capacity and guest-walk cost scenarios.
- Compare the hybrid and historical policies using business outcomes.

E. Figure Placeholder Example

Figures are essential for visualizing results. Below is an example of a figure placeholder, which can be adapted to include your project's visualizations.

These examples demonstrate how to format common elements in LaTeX while adhering to IEEE guidelines. Adapt these structures to present Your Business Problem's data, models, or results effectively.

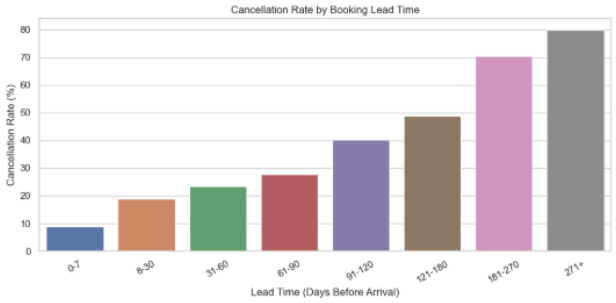


Fig. 2. Example Visualization of Model Predictions

VIII. Reproducibility

Reproducibility is ensured by sharing the dataset (from Kaggle), code, and methods. The workflow covers data validation, feature engineering, model training, calibration, and simulation.

To replicate results, obtain the dataset, install dependencies, and run the pipeline as described. The repository includes all scripts, configurations, and assumptions. Maintain the chronological split to avoid data leakage.

The business simulation uses scenario-based assumptions for capacity and costs. Document the nine evaluation scenarios and policy weighting (55% calibrated ML, 45% historical rates) to reproduce business results.

A. AI Use Declaration

AI tools, such as Your AI Tool (e.g., ChatGPT, Copilot, Grok, etc.), were used for Your AI Usage Description (e.g., generating initial code drafts or refining text). Ensure AI-generated content (text or code) constitutes less than 20% of the submission, as verified by Turnitin (https://www.turnitin.com), ensuring compliance with project guidelines.

IX. Conclusion

This study addressed hotel overbooking under cancellation uncertainty by employing machine learning and probability-based decision techniques. The findings indicate that the proposed hybrid approach, which integrates calibrated machine learning predictions with historical cancellation data, enhances overbooking decisions and profitability when appropriate capacity and guest-walk cost parameters are present. Future research may incorporate actual hotel capacity, observed guest-walk costs, and more detailed booking-time information to further refine the risk-adaptive decision system.

X. Acknowledgment (Optional)

We would like to express our sincere gratitude to Goa Institute of Management for providing us with the opportunity and resources to undertake this project. We are thankful to our faculty for their valuable guidance, feedback, and support throughout the project. We also acknowledge the publicly available Hotel Reservations Classification Dataset that supported our analysis. Finally, we thank our team members for their collaboration, commitment, and contributions toward completing this work.

References

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[3] The study uses the Hotel Reservations Classification Dataset published by Ahsan Raza on Kaggle https://www.kaggle.com/datasets/ahsan81/hotel-reservations-classification-dataset

[4] "From if to when: Timing hotel booking cancellations to improve demand management," International Journal of Hospitality Management, vol. 140, 2026, Art. no. 104858.